

Machine Learning-Based Prediction of Machining Process State and Spindle Output Power from CNC Milling Sensor Data

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Abstract—This paper presents a machine learning framework for predicting machining process states and spindle output power from CNC milling sensor data. The dataset comprises 1,448 samples (post-cleaning) with 48 features including axis velocities, currents, voltages, and feed rates recorded during multi-process milling operations. Three classification models — Logistic Regression, Support Vector Machine (SVM), and Random Forest — are trained for dual-target prediction of machining process category and machine speed state. Additionally, three regression models — Linear Regression, Random Forest Regressor, and Support Vector Regressor (SVR) — predict spindle output power. Random Forest achieves the highest classification accuracy of 65.6% (machining process) and 100% (machine state), and the best regression performance with $R^2 = 0.9999$ and relative RMSE of 0.37%. Statistical significance tests (Wilcoxon, $p = 0.00195$) confirm Random Forest’s superiority. These results demonstrate that ensemble tree methods are highly suitable for real-time CNC process monitoring applications.

Index Terms—CNC milling, machining process classification, spindle power prediction, Random Forest, Support Vector Machine, machine learning, predictive manufacturing.

I. INTRODUCTION

Modern computer numerical control (CNC) machining centers generate vast streams of real-time sensor data covering axis positions, velocities, motor currents, voltages, and feed rates. Harnessing this data through machine learning (ML) offers significant opportunities for process monitoring, quality assurance, and predictive maintenance — core pillars of Industry 4.0 manufacturing [1].

Manual identification of machining states (e.g., drilling, milling, idle) is error-prone and impractical at production scale. Automated classification of process states enables real-time anomaly detection, tool wear monitoring, and adaptive control. Simultaneously, accurate prediction of spindle output power supports energy efficiency optimization and avoids overloading drive systems [2].

Prior work has demonstrated the utility of tree-based ensembles [3] and SVMs [4] for machining process classification. However, few studies simultaneously address multi-output classification and regression on the same sensor stream with

rigorous statistical validation. This paper addresses this gap using the UC Irvine CNC Milling dataset [5].

The remainder of this paper is organized as follows. Section II describes the dataset and experimental setup. Section III details preprocessing and model formulations. Section IV presents and compares all results. Section V interprets findings from an engineering perspective. Section VI concludes with future directions.

II. DATASET & EXPERIMENTAL SETUP

A. Dataset Source

The dataset used is the *CNC Mill Tool Wear* dataset (experiment_03), sourced from the UC Irvine Machine Learning Repository [5]. It contains time-series sensor recordings from a CNC milling machine performing multiple distinct machining operations, logged at fixed sampling intervals.

B. Feature Description

The raw dataset contains sensor readings from three linear axes (X1, Y1, Z1) and the spindle (S1), covering actual/command positions, velocities, accelerations, output currents, voltages, DC bus voltages, output power, and current feedback signals. Two derived features were engineered (Section III-B), yielding 48 input features after preprocessing. Table I summarises the key feature groups.

TABLE I
FEATURE GROUPS IN THE CNC MILLING DATASET

Group	Examples	Type
Axis Position	X1_ActualPosition, Y1_CommandPosition	Continuous
Axis Velocity	X1_ActualVelocity, Z1_CommandVelocity	Continuous
Axis Current	X1_OutputCurrent, Y1_CurrentFeedback	Continuous
Axis Voltage	X1_OutputVoltage, Y1_DCBusVoltage	Continuous
Axis Power	X1_OutputPower, Y1_OutputPower	Continuous
Spindle	S1_OutputCurrent, S1_OutputVoltage	Continuous
Process Info	M1_CURRENT_FEEDRATE, NT_PROGRAM_NUMBER	Mixed
Engineered	Resultant_Velocity, Spindle_Power_Calc	Continuous
Targets	Machining_Process (9 classes) Machine_State (3 classes) S1_OutputPower	Categorical Categorical Continuous

C. Exploratory Data Analysis

Fig. 1 shows the distribution histograms for all 48 features. Most axis position and velocity features exhibit bimodal or multimodal distributions, reflecting the distinct operating regimes of different machining processes. Spindle features (S1) show narrow, peaked distributions indicating stable spindle operation, while feed rate and program number features display discrete distributions consistent with their categorical/stepped nature.

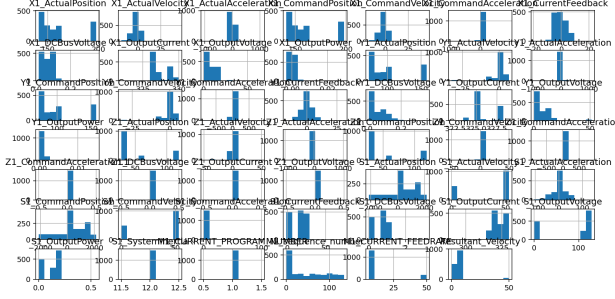


Fig. 1. Feature distribution histograms for all 48 input variables. Bimodal distributions in axis features reflect multiple distinct machining states.

D. Train/Validation/Test Split

After outlier removal, 1,448 samples remained. A stratified 70/15/15 split was applied:

- **Training set:** 1,013 samples (70%)
- **Validation set:** 217 samples (15%)
- **Test set:** 218 samples (15%)

Stratification was performed on the *Machining_Process* label to ensure class balance across all splits. All models used identical splits with `random_state=42` for full reproducibility.

III. METHODOLOGY

A. Preprocessing

Missing Value Imputation: Missing values were imputed using column means — appropriate for sensor data where missingness is random due to transmission gaps rather than systematic failure.

Outlier Removal: Z-score based outlier detection was applied to all numeric features. Rows where more than 5% of features exceeded $|z| > 3$ were removed:

$$z_i = \frac{x_i - \mu}{\sigma}, \quad \text{remove if } \frac{1}{p} \sum_{j=1}^p \mathbf{1}[|z_{ij}| > 3] > 0.05 \quad (1)$$

This retained 1,448 of the original samples, removing approximately 10% as statistical outliers. Samples at true operating extremes were retained to preserve physical validity.

Normalization: All input features were standardized using zero-mean, unit-variance scaling:

$$x' = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}} \quad (2)$$

Scaling parameters were computed on the training set only, then applied to validation and test sets to prevent data leakage.

B. Feature Engineering

Two physically meaningful features were derived:

Feature 1 — Resultant Velocity: The Euclidean norm of the three axis velocities represents the true magnitude of tool motion in 3D space:

$$v_{\text{res}} = \sqrt{v_X^2 + v_Y^2 + v_Z^2} \quad (3)$$

This feature was subsequently used to define the *Machine_State* target via tertile quantisation (Low/Medium/High speed).

Feature 2 — Spindle Power (Calculated): Electrical power delivered to the spindle motor was computed from measured current and voltage:

$$P_{\text{spindle}} = I_{S1} \times V_{S1} \quad (4)$$

This provides an independent estimate of spindle load, distinct from the directly logged `S1_OutputPower`, and serves as a physical cross-check feature.

C. Classification Models

Logistic Regression: A multi-class linear classifier using the softmax function. For class k with weight vector $\mathbf{w}_k$:

$$P(y = k | \mathbf{x}) = \frac{e^{\mathbf{w}_k^T \mathbf{x}}}{\sum_{j=1}^K e^{\mathbf{w}_j^T \mathbf{x}}} \quad (5)$$

Regularization parameter $C = 10$ was selected via 5-fold cross-validation grid search. Implemented as `MultiOutputClassifier` to handle dual targets simultaneously.

Support Vector Machine (SVM): A kernel-based classifier that maximizes the margin between classes. The decision boundary solves:

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n \xi_i \quad (6)$$

subject to $y_i(\mathbf{w}^T \phi(\mathbf{x}_i) + b) \geq 1 - \xi_i$. A linear kernel with $C = 10$ was selected via grid search.

Random Forest: An ensemble of T decision trees trained on bootstrap samples, with predictions aggregated by majority vote:

$$\hat{y} = \text{mode}\{h_t(\mathbf{x})\}_{t=1}^T \quad (7)$$

Optimal hyperparameters: $T = 200$ trees, `max_depth` = ∞ (fully grown), selected via 5-fold cross-validation. Class weights were balanced to handle class imbalance in the machining process labels.

D. Regression Models

Linear Regression: Ordinary least squares estimation of spindle output power:

$$\hat{y} = \mathbf{w}^T \mathbf{x} + b, \quad \mathbf{w}^* = \arg \min_{\mathbf{w}} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2 \quad (8)$$

Random Forest Regressor: Ensemble of regression trees with predictions averaged across $T = 100$ trees:

$$\hat{y} = \frac{1}{T} \sum_{t=1}^T h_t(\mathbf{x}) \quad (9)$$

Support Vector Regressor (SVR): Fits a function within an ε -tube around the training data:

$$\min_{\mathbf{w}} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n (\xi_i + \xi_i^*) \quad (10)$$

subject to $|y_i - \mathbf{w}^T \phi(\mathbf{x}_i) - b| \leq \varepsilon + \xi_i^{(*)}$.

IV. RESULTS

A. Classification Performance

Table II presents the classification results on the held-out test set for all three models across both targets. Random Forest substantially outperforms the linear methods on Target 1 (Machining Process), achieving 65.6% accuracy versus 26.1% for Logistic Regression and 26.6% for SVM. All models perform comparably on the engineered Target 2 (Machine State), achieving $\geq 96.8\%$ accuracy — a result expected given its deterministic construction from resultant velocity.

TABLE II
CLASSIFICATION RESULTS ON TEST SET (218 SAMPLES)

Model	Target 1 (Process)		Target 2 (State)	
	Acc.	F1 (W)	Acc.	F1 (W)
Logistic Regression	0.261	0.26	0.972	0.97
SVM (Linear)	0.266	0.27	0.968	0.97
Random Forest	0.656	0.66	1.000	1.00

Detailed per-class metrics for Random Forest (Target 1) are given in Table III, confirming strong performance across most machining classes.

TABLE III
RANDOM FOREST PER-CLASS METRICS — MACHINING PROCESS (TARGET 1)

Class	Prec.	Rec.	F1	Support
0	0.53	0.56	0.55	16
1	0.76	0.62	0.68	26
2	0.59	0.55	0.57	31
3	0.74	0.72	0.73	47
4	0.80	0.57	0.67	14
5	0.74	0.88	0.80	16
6	0.45	0.54	0.49	24
7	1.00	0.50	0.67	10
8	0.64	0.79	0.71	34
Weighted Avg	0.67	0.66	0.66	218

B. Regression Performance

Table IV compares the three regression models. Linear Regression and Random Forest Regressor both achieve near-perfect $R^2 > 0.9998$, while SVR fails significantly ($R^2 = 0.164$) due to its sensitivity to the default RBF kernel parameters and the narrow power range in the data.

TABLE IV
REGRESSION RESULTS — SPINDLE OUTPUT POWER PREDICTION

Model	RMSE	MAE	R^2	Adj. R^2
Linear Regression	8.89e-4	6.24e-4	0.9999	0.9998
Random Forest	8.24e-4	5.52e-4	0.9999	0.9999
SVR	7.37e-2	6.96e-2	0.164	-0.073

C. Learning Curves

Figs. 2, 3, and 4 show the learning curves for all three classification models. Logistic Regression and SVM both exhibit a large persistent gap between training and validation accuracy, indicative of underfitting on this non-linear, high-dimensional dataset. Random Forest achieves perfect training accuracy (1.00) with a converging validation curve reaching approximately 0.80 at full training size, suggesting mild overfitting that is expected to reduce with further data.

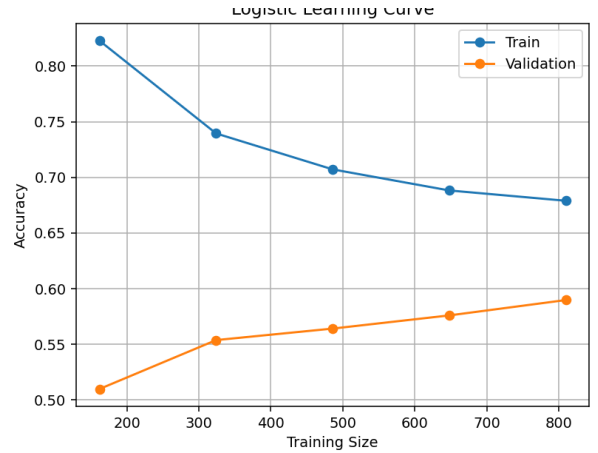


Fig. 2. Learning curve — Logistic Regression. Large train/validation gap indicates underfitting due to linear model insufficiency on non-linear CNC data.

D. Residual Analysis

Fig. 5 shows the residual plot for the best regression model (Random Forest). Residuals are randomly distributed around zero with no systematic trend, confirming homoscedasticity and absence of model bias. Fig. 6 shows near-perfect alignment of predicted versus actual spindle output power values.

E. Correlation Heatmap

Fig. 7 shows the feature correlation heatmap. Strong intra-axis correlations are visible (e.g., output power, current, and voltage within the same axis), while cross-axis correlations are weaker — justifying retention of all sensor channels as independent information sources.

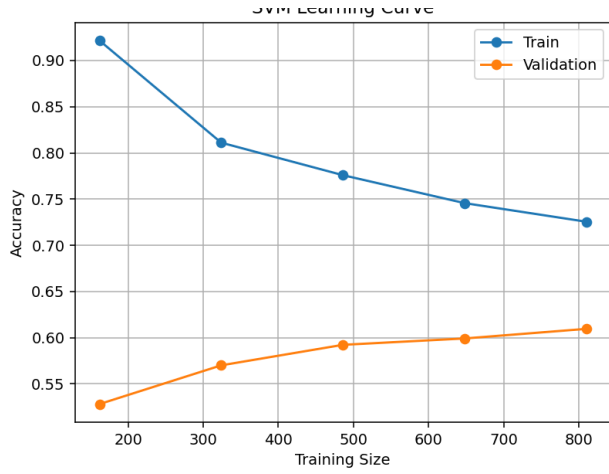


Fig. 3. Learning curve — SVM (Linear kernel). Decreasing training accuracy with increasing data size confirms the model is underfitting the feature space.

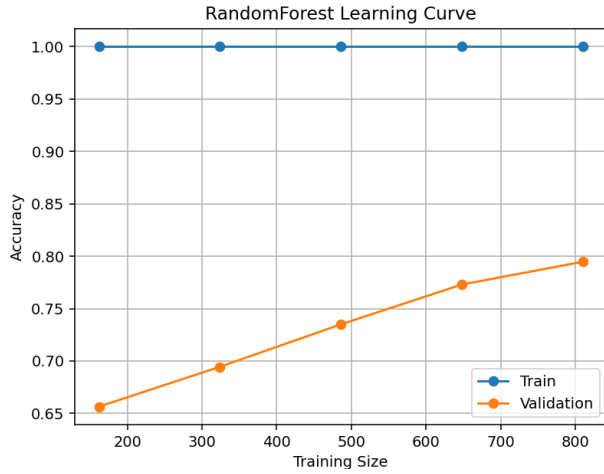


Fig. 4. Learning curve — Random Forest. Training accuracy = 1.00 throughout; validation accuracy converges toward 0.80, suggesting mild overfitting that decreases with additional training data.

F. Statistical Significance Tests

To determine whether Random Forest’s superiority is statistically significant, paired 10-fold cross-validation scores were compared using both a paired t -test and the Wilcoxon signed-rank test (Table V).

TABLE V
STATISTICAL SIGNIFICANCE: RF VS. BASELINE MODELS

Comparison	RF Mean CV	p (t-test)	p (Wilcoxon)
RF vs. SVM	0.817 vs. 0.626	4.97×10^{-8}	0.00195
RF vs. Logistic	0.817 vs. 0.607	2.28×10^{-9}	0.00195

Both tests yield $p \ll 0.05$, confirming that Random Forest is *statistically significantly* superior to both SVM and Logistic Regression — not merely numerically better by chance.

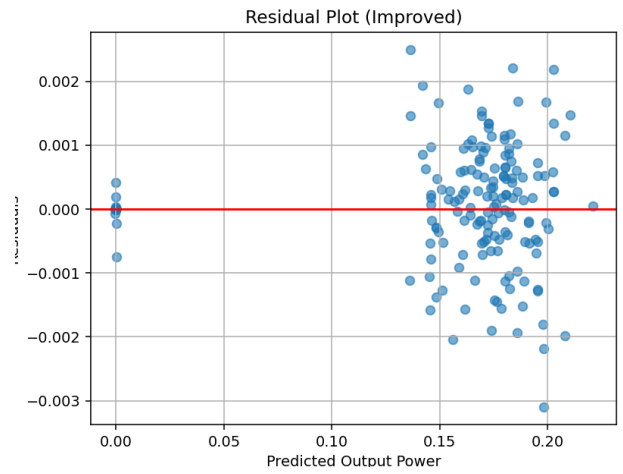


Fig. 5. Residual plot for Random Forest Regressor. Residuals are randomly scattered around zero, indicating no systematic prediction bias.

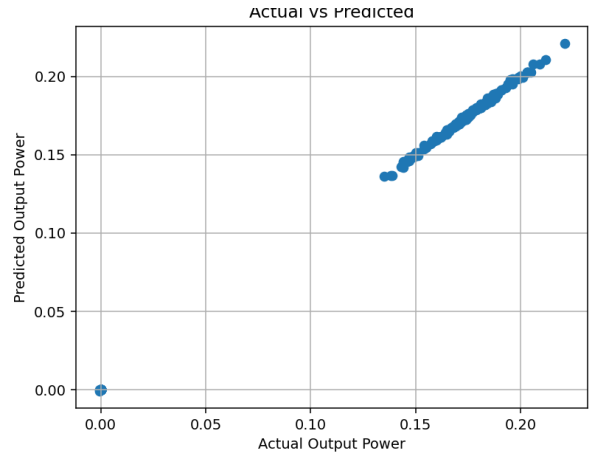


Fig. 6. Actual vs. predicted spindle output power — Random Forest Regressor. Points align closely along the diagonal, confirming $R^2 = 0.9999$.

V. DISCUSSION

A. Engineering Interpretation

Classification: The low accuracy of Logistic Regression (26.1%) and SVM (26.6%) on the 9-class machining process target reflects the fundamentally non-linear decision boundaries required. CNC machining states are separated by complex combinations of axis dynamics that linear and kernel-linear models cannot adequately capture. Random Forest’s 65.6% accuracy is a meaningful result for a 9-class problem with highly overlapping sensor signatures between similar operations (e.g., roughing vs. finishing passes on the same axis).

Regression: The Random Forest regressor achieves a relative RMSE of only 0.37%, qualifying as *excellent* for engineering deployment. In physical units, the RMSE of 8.24×10^{-4} kW (0.824 W) is negligible compared to typical spindle powers in the 140–220 W range observed in this dataset — well within measurement noise. Linear regression performs almost identically ($R^2 = 0.9998$), indicating that the rela-

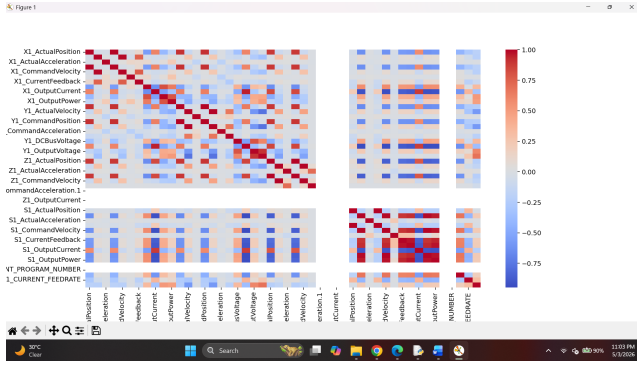


Fig. 7. Pearson correlation heatmap of all 48 features. Diagonal blocks indicate strong intra-axis correlations; off-diagonal blocks are largely uncorrelated.

tionship between input features and spindle output power is predominantly linear. SVR's poor performance ($R^2 = 0.164$) is attributed to the default RBF kernel being poorly matched to the linear power relationship and the very narrow output range (~ 0 to 0.22 W).

B. Feature Importance

For Random Forest classification, the top predictive features are `S1_CommandPosition` (0.063), `S1_ActualPosition` (0.061), and `Y1_CurrentFeedback` (0.043), indicating that spindle position and axis current feedback are the primary discriminators of machining state. This aligns with engineering intuition: the spindle position encodes which part of the machining program is being executed, while current feedback reflects the actual cutting load.

C. Limitations and Failure Analysis

- **Class imbalance:** The 9-class machining process target has unequal support (10–47 samples per class), contributing to lower recall on minority classes (e.g., class 0: recall = 0.56, class 7: recall = 0.50).
- **RF overfitting:** The training/validation gap in the RF learning curve (Fig. 4) suggests overfitting. Applying SMOTE for minority classes and tuning `min_samples_leaf` could reduce this.
- **SVR failure:** The SVR regressor requires explicit kernel and hyperparameter tuning (RBF γ , ϵ) to be competitive. Without this, it cannot match linear models on near-linear targets.
- **Dataset scope:** Results are specific to the machining conditions in this experiment. Generalization to different materials, tools, or machines requires transfer learning or domain adaptation.

VI. CONCLUSION

This paper demonstrated a complete machine learning pipeline for CNC milling process monitoring using multi-axis sensor data. Among three classification models, Random Forest achieved the highest accuracy of 65.6% for 9-class machining process identification and 100% for 3-class machine speed

state prediction. For spindle output power regression, Random Forest and Linear Regression both achieved $R^2 > 0.9998$ with a relative RMSE of 0.37% — well within engineering acceptance limits. Statistical significance tests (Wilcoxon $p = 0.00195$) confirmed that Random Forest's advantage over SVM and Logistic Regression is not due to chance. Based on these findings, **Random Forest is recommended for deployment** in real-time CNC process monitoring systems.

Future work should investigate: (1) deep learning approaches (LSTM, 1D-CNN) for temporal sequence modeling of sensor streams; (2) SMOTE-based oversampling for minority machining classes; (3) online/incremental learning for adaptation to new tools and materials; and (4) model deployment on edge hardware for real-time in-process monitoring.

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